

HINTERRHEIN, SwitzerlandWMO: **067881**Lat: **46.524N**Lon: **9.184E**Elev: **5308**StdP: **12.09**Time Zone: **1.00 (EUC)**Period: **01-11**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.7	(c) 1.3	(d) -13.4	(e) 3.2	(f) 6.8	(g) -9.1	(h) 4.1	(i) 5.0	(j) 33.7	(k) 30.7	(l) 24.9	(m) 29.8	(n) 2.1	(o) 230	(p) 0.722

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	72.8	56.9	70.2	56.2	67.7	55.4	59.4	68.6	57.8	66.8	56.4	65.2	8.3	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.4	79.8	64.0	54.0	75.6	62.3	52.7	72.0	60.6	28.5	68.9	27.4	67.0	26.4	65.2	63.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 24.2	(b) 19.9	(c) 16.7	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.9	19.9	21.9	28.4	37.4	44.9	52.1	54.9	53.8	47.2	41.3	31.1	21.3
	DBStd	14.29	8.38	7.57	7.75	6.17	5.67	5.90	5.00	5.01	5.57	6.65	8.31	9.09
	HDD50	4872	933	787	669	377	176	47	16	22	115	274	566	890
	HDD65	9879	1398	1207	1134	827	624	387	314	347	534	736	1016	1355
	CDD50	467	0	0	0	0	17	110	167	140	30	3	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	37	0	0	0	0	0	6	6	25	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.8	6.3	6.8	7.5	7.7	7.2	6.1	6.0	6.2	6.7	7.8	7.0	6.5
Precipitation	PrecAvg	52.0	2.7	2.5	2.7	3.9	4.9	5.4	5.3	5.9	5.1	5.0	5.6	3.0
	PrecMax	70.7	6.2	7.3	6.9	11.3	14.2	10.7	9.0	8.8	12.5	17.1	22.9	6.3
	PrecMin	32.6	0.7	0.6	0.4	0.6	1.5	1.5	2.5	0.9	1.0	0.6	1.0	0.2
	PrecStd	9.4	1.5	1.8	1.7	2.4	2.9	2.6	1.7	1.8	3.4	3.3	4.8	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.9	43.3	49.9	61.6	67.5	75.0	74.8	77.5	68.8	63.1	53.4	41.1
		MCWB	33.5	34.3	39.1	45.7	52.4	57.0	57.7	56.3	54.5	48.5	43.8	34.4
	2%	DB	37.7	38.8	45.4	56.7	63.1	71.9	72.2	72.5	64.5	58.1	48.4	38.6
		MCWB	32.5	32.5	36.9	44.0	49.3	56.3	57.2	56.9	51.9	46.0	41.4	34.1
	5%	DB	35.0	35.9	42.6	51.9	60.2	69.2	69.7	68.9	61.1	54.9	44.2	36.7
		MCWB	31.3	31.1	35.4	41.6	48.2	55.4	56.8	55.8	50.8	45.7	38.4	33.0
	10%	DB	32.7	33.2	40.0	47.9	56.2	65.8	67.3	65.1	57.8	52.1	41.5	34.2
		MCWB	29.5	28.8	34.2	39.6	47.0	54.8	56.0	55.1	49.8	46.0	37.7	31.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.6	36.0	40.1	47.5	53.8	60.7	61.1	61.1	56.8	50.9	46.3	37.3
		MCDB	40.1	41.1	46.9	58.1	65.0	69.7	69.9	71.5	65.9	57.2	51.3	39.3
	2%	WB	33.5	33.4	38.0	44.7	51.4	58.5	59.5	58.5	54.1	49.5	41.9	35.3
		MCDB	36.0	37.7	43.1	54.8	60.7	68.5	68.7	67.1	60.9	54.7	47.0	37.6
	5%	WB	31.6	31.3	36.5	42.6	49.5	56.7	58.1	57.0	52.3	48.0	39.7	33.3
		MCDB	34.7	35.0	41.4	50.4	57.7	66.5	66.9	65.6	57.9	52.8	43.2	35.8
	10%	WB	29.5	29.6	34.8	40.5	47.6	54.8	56.7	55.5	50.7	46.1	38.1	31.4
		MCDB	32.8	32.9	39.3	46.9	54.6	64.3	65.1	63.7	56.3	51.0	41.2	34.2
Mean Daily Temperature Range	5% DB	MDBR	15.9	17.8	16.9	15.1	16.2	18.6	19.2	17.5	16.5	15.6	12.8	13.8
		MCDBR	17.0	20.1	19.9	22.9	24.2	26.8	24.8	26.6	24.5	23.4	17.1	13.8
		MCWBR	12.7	15.1	12.9	12.0	12.7	13.3	12.3	13.0	13.2	12.9	10.8	10.0
	5% WB	MCDBR	15.1	16.9	16.7	21.1	20.3	23.6	22.1	22.4	19.7	16.6	14.2	11.0
		MCWBR	11.6	12.9	10.9	11.5	11.4	12.4	11.8	11.8	11.0	10.0	9.6	8.8
Clear-Sky Solar Irradiance	taub	0.237	0.261	0.303	0.348	0.363	0.372	0.367	0.359	0.338	0.312	0.269	0.235	
	taud	2.331	2.267	2.206	2.197	2.295	2.315	2.330	2.369	2.415	2.463	2.450	2.366	
	Ebn at Noon	279	292	292	287	286	283	283	281	278	270	265	268	
	Edh at Noon	23	30	38	43	40	40	38	36	31	25	21	20	
All-Sky Solar Radiation	RadAvg	509	817	1202	1416	1514	1781	1780	1544	1203	800	496	424	
	RadStd	63	95	174	230	152	159	155	137	118	94	74	55	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (120 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+1.30	+0.74	+0.56	N/A	N/A	+106	+29

Nomenclature: See separate page